Exercises

Problem 3.1
Prove Eqs. (3.49) and (3.50) by using the Campbell identity.

Problem 3.2
Prove the conservation of the commutation relation for rotated quadratures, Eq. (3.55).

Problem 3.3: Mean and variance under displacements
Consider a quantum harmonic ocillator in state $|\psi\rangle$. Show that a displacement $|\psi\rangle \to \hat{D}(\alpha)|\psi\rangle$ transforms the expectation values of $\hat{x}$ and $\hat{p}$ as

$$\langle\hat{x}\rangle_\psi = \langle\psi|\hat{x}|\psi\rangle \to \langle\hat{x}\rangle_\psi + \sqrt{2}\,\mathrm{Re}(\alpha),$$
$$\langle\hat{p}\rangle_\psi = \langle\psi|\hat{p}|\psi\rangle \to \langle\hat{p}\rangle_\psi + \sqrt{2}\,\mathrm{Im}(\alpha).$$

Show that the variances $\langle\Delta^2\hat{x}\rangle$ and $\langle\Delta^2\hat{p}\rangle$ remain unchanged.

Problem 3.4: Reality of the Wigner function
Show that the Wigner function as defined by Eq. (3.80) is real.

Problem 3.5: Wigner function marginals
Show that for pure states, the marginals of the Wigner function equal the probability densities given by the $\hat{x}$- and $\hat{p}$-space wave function as in Eq. (3.83).

Problem 3.6: Single-mode squeezing transformation
Show that the annihilation operator transforms under single-mode squeezing as given by Eq. (3.97).

Problem 3.7: Two-mode squeezing transformation
Show that the annihilation operators for a pair of modes transform under two-mode squeezing as given by Eqs. (3.109) and (3.110).

Problem 3.8: Beam splitter between coherent states
Prove Eq. (3.141), showing that products of coherent states remain products when mixed on beam splitters.

Problem 3.9: Implementing displacements
Show that the construction described in Sec. 3.2.2 and Fig. 3.13 realises a displacement for arbitrary input states. *Hint: you can use the P-representation.*

Problem 3.10: Coherent state subject to loss
A coherent state $|\alpha\rangle$ is sent through a lossy optical channel with transmissivity η. What is the output state? What is the mean number of photons detected from a coherent state $|\alpha\rangle$ by a photon counter with efficiency η (and no dark counts)?

Problem 3.11: Quadratures and beam splitters

Two modes A and B are mixed on a beam splitter with transmissivity η as in Fig. 3.11. Show that the quadrature operators transform as

$$\hat{U}_{BS}(\eta)\,\hat{x}_A\,\hat{U}_{BS}^\dagger(\eta) = \sqrt{\eta}\,\hat{x}_A + \sqrt{1-\eta}\,\hat{x}_B, \tag{3.183}$$

$$\hat{U}_{BS}(\eta)\,\hat{p}_A\,\hat{U}_{BS}^\dagger(\eta) = \sqrt{\eta}\,\hat{p}_A + \sqrt{1-\eta}\,\hat{p}_B, \tag{3.184}$$

$$\hat{U}_{BS}(\eta)\,\hat{x}_B\,\hat{U}_{BS}^\dagger(\eta) = \sqrt{1-\eta}\,\hat{x}_A - \sqrt{\eta}\,\hat{x}_B, \tag{3.185}$$

$$\hat{U}_{BS}(\eta)\,\hat{p}_B\,\hat{U}_{BS}^\dagger(\eta) = \sqrt{1-\eta}\,\hat{p}_A - \sqrt{\eta}\,\hat{p}_B. \tag{3.186}$$

Conclude that for any θ, the rotated quadrature transforms the same way, i.e.

$$\hat{U}_{BS}(\eta)\,\hat{x}_A(\theta)\,\hat{U}_{BS}^\dagger(\eta) = \sqrt{\eta}\,\hat{x}_A(\theta) + \sqrt{1-\eta}\,\hat{x}_B(\theta), \tag{3.187}$$

$$\hat{U}_{BS}(\eta)\,\hat{x}_B(\theta)\,\hat{U}_{BS}^\dagger(\eta) = \sqrt{1-\eta}\,\hat{x}_A(\theta) - \sqrt{\eta}\,\hat{x}_B(\theta). \tag{3.188}$$

Problem 3.12: Squeezing and loss

Consider that a single-mode squeezed vacuum state is transmitted through a lossy optical channel with transmissivity η and then measured with perfect homodyne detection (or equivalently, the input state is measured with a homodyne detector with efficiency η). You can take the squeezing to be in the $\hat{x}$ quadrature and assume that the detector also measures $\hat{x}$. What is the variance in the measurement outcome (i.e. how much squeezing will be measured)?

Problem 3.13: Lossy bucket detector POVM

Show that the POVM elements corresponding to a measurement by a bucket detector (i.e. a non-number-resolving single-photon detector) are given by

$$\hat{E}_0 = \sum_{n=0}^{\infty}(1-\eta)^n|n\rangle\langle n|, \qquad \text{and} \qquad \hat{E}_1 = \mathbb{1} - \hat{E}_0, \tag{3.189}$$

where $\hat{E}_0$ and $\hat{E}_1$ correspond to the 'no click' and 'click' outcomes, respectively.